

The Observer Is The Observed with Karl Friston and Chris Fields

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in our culture we believe that reality means physical things question is not whether the world is real or not the question is what is reality in this Theory we are delighted to have Professor Karl Friston it's a theoretical neuroscientist and an authority on brain Imaging with Dr Chris Fields a philosopher and an independent scientist into radical physics and cognitive neuroscience the topic of our discussion today is The Observer versus the observed welcome Professor Friston welcome Dr Phil thank you thank you so anything interesting going for both of you have you been anywhere lately beside work Professor Christine no I'm afraid not I I'm just yeah but my work is a holiday so anything interesting going for you this year oh no I've had my uh stepmother visiting for a week so we've had a bit of a family week this last week today at your Chinese New Year yeah today this year is a rabbit year our calendar just changed today it's the fifth day of it tell us briefly how your career being my career yeah yeah um so I started as a medical trainee um largely because I wanted to do what was called in those days mathematical psychology but my careers advisor mistakenly thought that I had to be a psychiatrist first so I went to University dutifully to study Psychiatry but was able um to do physics as an undergraduate so I did Natural Sciences that included psychology and physics and then became a medic and a psychiatrist and then at the earliest opportunity returned to um the kind of research that I wanted to do as an adolescent which was basically mathematical psychology so this book Professor Christine you said you're going to write about something about an angel I heard a talk about you talking about a book that you want to write so if you're going to write a book about an angel so would it be your philosophy or would it be something different could you talk on that please yes um yes the the idea the word or the phrase or the um the concept Angel was something that arose uh during my early psychiatric training from um what was called in fact is called T group therapy so part of my training was to live in a Therapeutic Community um with about 30 people with chronic schizophrenia so I spent about two years of

my life living in this somewhat isolated Community with psychiatric nurses and doctors and people with schizophrenia and with every um week we would have several group sessions and in these sessions all sorts of wonderful things were talked about and one of the more memorable themes was this notion from a very high functioning and very endearing um uh young man with schizophrenia who just came up with this notion of Angel and just struck me as such an interesting conjunction of things that we don't normally associate you know a nice expression of what's called night's move or lateral thinking and in the Psychopathology of schizophrenia but also reflecting the creativity um of somebody who's able to think outside normal bounds and to transcend the normal syntax and semantics in the language that we usually share but I've always remembered that instance um and the phrase aim and I thought to myself when I get older I will write a book or the Angels but as I get older I realize there is no time to write books so I'm afraid I won't be writing a book called Angel it would be thank you Chris you begin as an experimental physicist could you tell us more uh well I beginning graduate school working at a small accelerator doing nuclear structure physics and uh partway through that era I by for various reasons switched into philosophy and took up with people doing artificial intelligence shortly after that ended up applying some of that work in work for the Human Genome Project so I spent a good deal of my career in the human Human Genome Project so I have kind of a mixed bag from past so it used to find time to be a volunteer as a firefighter right Chris a small village in New Mexico and the fire department was one of the things that held our small village with a very diverse population together and so I spent a few years doing that oh it was very interesting dealt with a lot of car wrecks on icy roads in the middle of the winter thank you so two great minds let's get serious now so when we say reality in our culture we say reality is mean physical thing so shall we go to to call first what did you mean when we say reality I'm trying to um when you say what do we mean I would imagine that um from a philosophical point of view we're talking about metaphysics so yesterday I was given a briefing tutorial on metaphysics um and from the point of view of the free energy principle which is the lens through which I usually use uh you view many things and reality I guess would correspond to um the states or the configuration of things beyond our sensorium beyond our Markov blanket beyond our holographic screen and so they would be the putative causes of the things that we can read from nature or that we um would think about in terms of the sensory Impressions and our sensory epithelia or as I'm hoping Chris will talk about um the the sensory bits the readable bits on some screen or Veil or boundary or holographic um screen that separates me from the rest of the universe so reality on that vehicle is just the X the

exterior and strictly speaking from my point of view uh that that reality is something which is constructed on the inside um and uh can only kind of insight as being our awareness our experience yeah constructed inside yeah yeah so so the the I I imagine that um you will want to talk about um this issue from the perspective of the free energy principle read in two ways read from the point of view of a classical physicists point of view and possibly also nowadays A cognitive neuroscientist point of view uh in the sense of things like predictive processing but there's another way of reading it which may be a more generic way which is the Quantum information theoretic way of understanding things uh that I'm hoping to learn um more about by when you start quizzing um so the classical uh the classical view um means that you are separating all states of Any Given Universe into um separate subsets with a particular partition that separates or individuates the internal States the interior of something from the exterior the external States and there are that separation is um defined in terms of a Markov blanket a blanket blanket outside so if my if I am um somewhere in my brain or my brain is is part of me then the internal States would be my neuronal States and my scent the sensory part of my Markov blanket would be those signals that correspond to sensory receptors say you know vision and hearing but also interception anything um that um reports or um signifies changes in the outside in the external states that that is available to me so when I talk about the inside I I mean the internal the bulk on the inside of say a brain the kind of thing that we study with brain Imaging for example or indeed make inferences about it as experimental psychologists using uh psychophysics um or other psychological tools to make a guess about what's going on on the inside of somebody's head thank you Chris uh well I certainly won't disagree with that uh I'll translate your question a bit to uh what are we talking about when we talk about the physical and here we can look back at the history of physics as a discipline uh which evolved from something that talked about uh ordinary objects like planets and cannonballs and such into something that now uh is formulated at least in many sectors of the discipline isn't it as an information theory that basically talks about information and information processing so in a sense the boundaries between physics and Computing are starting to dissolve in at least certain parts of physics Quantum cosmology for example so when we talk about the physical uh we're talking about something that can be characterized in terms of some sort of State space of something that has distinguishable configurations or States and so when we talk about the physical in a sense we're we're making some assumption about how we're going to characterize what can be distinguished and hence we're talking about information which is uh how what it means to distinguish something is to talk about information so I tend to think

about your question uh and and also Carl's answer in terms of a division of some universe as Carl was saying uh and some physicists refer to this as a Multiverse uh into two pieces of that exchange information by interacting by physically interacting and to ask what does that exchange of information look like how can we model that exchange of information and how can we model what the systems on both sides of this this division that we've stipulated how can we model what they do with the information so how does each agent or entity act on the other one and a natural way to think about what they do with the information is is clearly the theory of computation uh they're processing the information in some way so it's very natural to think of this this physical interaction as we would call it as a communication between two entities who are each receiving information from the other doing something with it uh which we can think of as a decision process or a computational process and then they're sending a message back or they're acting on the other system in some in some physical way all these ways of of talking are formally completely equivalent so one can bring the language of communication in the language of computing to bear on on any physical interaction so uh this naturally raises the question if we think of of each system as some kind of information processor how do they structure the input that they receive from the other system and uh one way to to think about that question is to ask uh can they detect a correlated patterns in the input that they get from the other system so if we think of this in communication language can they detect words or sentences or other meaningful uh subsets in the communications they get from the if we think about it in terms of a sense like vision uh we would ask can they see objects that hold together over time being able to detect such structure in the information that they receive that each receives from the other system is necessary in order to do any sort of prediction so when Carl talks about the fep it's a least action principle that has to do with minimizing uncertainty about what one's interaction partner will do next being able to do predictions requires being able to to capture some kinds of uh regularity is in the in the in the input coming from the other system so it has to be able to locate parts of that signal that have relatively low entropy in other words some sort of relative stability over time so if we pursue this line of thinking of were very quickly led to viewing what we call ordinary objects like uh planets or billiard balls or computers or each other as constructs as Carl put it because they are as of low entropy in the signals that we're receiving from whatever it is that we're interacting with that we see uh only via the messages that it sends to us or only via the actions that it takes on us uh and so uh what is often called reality uh what what we see all around us and hear all around us is is in that sense a construct it's something that we have to be smart enough to put

together using whatever information processing facilities that we have uh including of course those in our brains and that that leads us of course almost immediately to ask a different question which is lots of the things we see around us uh look for all the world like other agents rather like us uh not just other humans but all other organisms and so we can ask how did they see their world yeah what what is an object for them uh how do they view uh what their environment is doing to them all the time so one is quickly drawn into a kind of ethology if you will uh concerned with how other organisms process what they're getting from their environments thank you Carl do you have anything else before we move on lots of things come to mind I I think it might be quite useful um just to paraphrase some of the terms that Chris was using um in the language of inference and Bayesian inference in particular and just to put a bit of intuition for those people who are not as um fluent in Quantum information theories as as other people are yeah so you know words like computation information processing uh I would also say measurement and be written as Chris nicely expressed you know what natural way do we have of describing the the sense that is made of this information that is arriving um on our sensory epithelia or on our um a Markov blanket or um in the um the quantum formulation on the holographic screen um one way of talking about information processing is just to cast it as belief updating where you've introduced now the notion of a probability distribution so if you just think about reality being a particular dynamics of probability distributions very much like quantum um then um what you are talking about when you talk about information processing is the change in some probabilistic representation on the inside in the brain for example that is induced or consequent Upon A Certain set of then three inputs and this belief updating can be read in many different ways you know that you from a purely mathematical in terms of movement on a statistical manifold every point in these special spaces these special manifolds corresponds to a certain belief where a belief is just a probability distribution over reality which are the external States so you never see them but you can make inferences about them so for me information processing is just a movement on a statistical manifold and then Chris also mentioned the word structure you know how do we make sense of how do we detect those regimes of low entropy those bits of information that have some causal structure so in things like radical constructivism and indeed uh Bayesian modeling this would be structure learning it's getting the frames of reference right so that you are using the right kind of manifold to do your information processing by moving around on that manifold so that you can make sense of the data looking for symmetries and invariances um that would correspond to again as Chris noted objects have certain invariances you know

that they have certain features that are invariant to their location and their location may be your location as a construct doesn't necessarily depend upon the object that that is at that location so keying apart finding the conditional independencies is one way of looking at getting these frames of reference fit for purpose to do the kind of information for us to say and the best kind is the one that makes this makes sense of all these sensory signals um sometimes you can you well some people refer to that as sort of being Bayes optimal being optimal in the sense of finding that structure that frame of reference on the inside that upon which this belief amputating and information processing plays out that renders all of those input as likely as possible and that's you know it is from my point of view it's as simple as that um so you can score this as likely as possible in terms of things like Bayesian model evidence or um the complement or the inverse of that things like um variational free energy or prediction error surprise or information Theory or more in terms of psychologist's surprise so the best stuff is the stuff which are the best sense making is a sense making that renders you the least surprised about seeing this pattern of sensory Impressions from reality out there you'll never know what reality out there is doing but uh at least you can make the best guess of this by finding the right kinds of structures what Chris has done though he's taken us right into uh the much more glorious realm of two sense making systems that are talking to each other and further to that the there's some isomorphism there so you imagine that's two highly sophisticated and evolved structures that are passing messages across their with a shared screen if you like or a shared Markov bag that's a shared separation um and you know all the interesting issues that arise from that that can be cast in terms of um say generalized synchrony from the point of view of classical dynamical systems theory from the point of view entanglement in in Quantum three months but the other interesting um perspective that uh Chris has uh has his new month of the argument is that to get this minimum surprising maximally predictable um State of Affairs then there has to be a shared narrative on both sides a shared structure to establish this synchrony which could be read in terms of sense making and information processing as basically resolving the hermeneutics problem um and of course how do we do that well it's with language so uh like you know and if you generalize that notion of language then all communication is just good belief updating in the service of expanding something out there that actually shares some reference frame or some to share some narrative or World model or the statistical manifold with me and I have to learn uh I have to learn about you they have to speak the same language and we converge upon the same kind of structure just to make sure that the sensations that we receive are as likely and as predictable as possible thank you so last

week we have Don and Levin on our show so don't reduce reality to a network of conscious agents then he then went one step further that was he reduced all this multiplicity and diversity of Pontius aging into a single consciousness because they're all they're all our conscious experiences so the Consciousness is the common factors of it so my question is at what level one can be described as having a conscious experience though can we go to Chris I take a fairly radical or deflationary or one could call it various things view of this oh yeah reason in this way that what we call experience is always uh determinate or classical uh it's it's what it's what in a sense we mean by classical information uh you know we we don't see Schrodinger's cat as being dead and alive we see Schrodinger's cat as being either one or the other and uh we don't see people as being located both here and there we see them located in one place and saying one thing and so on so that's that's classical and from a fairly strict kind of Quantum information theoretic perspective the only place that one expects to see classical information is on a boundary between two interacting systems if we think of classical information as only encoded on boundaries and if we think of experience uh as classical information then it's fairly natural to think that experience is what's happening on a boundary and because that's where the classical information is it's not anywhere else in this very kind of strict Quantum theoretic point of view in that case one can think of the universe if you will is this big system of which we can describe using at least a quantum formalism and one can think of slicing it in various ways and so slice it in some arbitrary way you've created a boundary now that boundary may not be stable if the if the interaction across that boundary is very strong then the boundaries not going to pick out two systems that one can talk about as if they're different oh and formally if if the interaction is strong enough that the entanglement entropy across that boundary is large uh then those systems don't really have separate identities they're too correlated to be considered to be just in exchanging classical information but if the interaction across the boundary that you've picked is fairly weak uh then the two systems are more or less independent of each other and you can think of them as separate entities that have their own States and that are exchanging some relatively small amount of information with each other compared to the amount of information that they've got inside and that's that's the place where the free energy principle is helpful because it depends on this conditional independence of one system from another so now if if you've if you've picked a slice like that when you you slice the universe up then that boundary is going to encode some classical information and I see no reason not to think of all of that classical information is experiential or think of it in experiential terms and we're very familiar with that in our own case and we're reasonably

familiar with it in the case of animals like us and we have essentially no familiarity with familiarity at all with it in the case of other systems but I don't see any reason not to think of it that way for arbitrary systems that interact sufficiently weakly with their environments so I would be uh either bold or crazy and just say we should think of experience in general as the classical information that's encoded on boundaries where the boundaries can be arbitrary but the amount of classical information they encode depends on uh whether they separate systems that are uh separable or Mutual mutually conditionally and dependent and the way that Carl was mentioned I agree with that absolutely entirely um one nice thing about um the quantum formulation of the fep is its scale-free nature and um one could read what Chris just said as um saying that there are Markov blankets and many many different scales and um certain kinds of systems that are enshrouded in a particular blanket or boundary um will certainly have it's only be described as doing belief updating and um making sense of data experiencing the sensory inputs in in the um in the formulation of the Chris just offered and as such I think it's perfectly permissible to say that there can be um a universe of collective consciousnesses um just in virtue of the fact that you know as we've just heard there are many many ways of carving up conditional independencies and developing Markov blankets at different scale so I see no mathematical problem um with having distributed cognition and you're a mathematical form of collective Consciousness in which there is a minimization of the joint free energy of multiple Markov blankets that are already founded by a Markov blanket defying that particular community of sense making agents for example those sanctions could be people but I would also so maybe just think about um the brain as a very carefully structured um Ensemble of Markov blanketed bounded um systems neural populations specialized cortical areas that themselves comprise uh lots of delicately structured Markov blankets at the level of single neurons and then within a single neuron axonal dendritic processes a ride down to to the molecular level so viewed it from the scale free perspective from Gaia down to down to um the you know the molecular scale um then if the same kinds of principles apply and there's I suppose can be read in terms of sense making and sentience of a basal um sort of elementary form then it can be no other way that that Consciousness cannot be tied to one privilege knock off blanket so I think that that's an interesting observation um and um it it does strike me though that you may be asking um there may be a difference between experience and qualitative experience so you may be asking about qualia uh and you know does every yeah yeah um does every um individuated uh boundary system um are you licensed to suppose that it has the capacity to experience qualitatively in you know in a phenomenological sense um I think

that's you know so we're touching on things like pan psychism now um and I think there are two ways that you can you can read this the first is yes obviously um there is a basal sense making a basal distributed cognition a distributed Consciousness you know that is evident in the in any kind of communicating sparsity Capital system from you know multiple molecules to multiple cells to organs in my body to to institutions or you know bodies of people um or you could argue well yes that's fine but are there any special kinds of sense making that um would give you the um the qualitative experience that people talk about in philosophy I would suggest that that that there may be um and you know um after many years of conversing with uh people in the field I think one key distinction between the kind of experience that is generic to any self-evidencing uh system that can be individuated from its milieu it's external milieus echinical environment is the capacity to select to act on the world out there in such a way that it has consequences for other sensory impressions so if I was limiting myself to the brain for example and I was thinking like a psychologist this would be attention mental action actively selecting what kinds of sensory Impressions make a difference to my information processing and my belief updating um if um I was a motor physiologist I'd be talking about being able to move say my eyes and look at that so seeing becomes looking at um if I was a physiologist I would make a distinction between the things that I have causative experience of and things that I don't so I when I breathe I can attend to and also using my muscles exert volitional control of my respiration but I can't do that with my heart so I'm not aware I don't have a phonology of my my cardiac cycle that I do of my respiratory title so I think that the the special kinds of actions the acting the the other way if you like the other direction of travel through the boundaries or in the other direction of travel in terms of information [Music] um channels that transcend or transgress these boundaries or or the Markov blanket which is writing to the boundary or then acting upon the other side which could be another person I think that may be an important kind of structure that that would be um possibly um necessary to actually have a qualitative experience as opposed to just an experience thank you so Chris it's it's an entanglement then between the two right through the user interface like Don saying so it's an entangle system between the two interactions then right is that what you're saying uh well entanglement is one of those words that has different meanings uh in different disciplines in in physics entanglement or quantum entanglement is uh defined as the state of non-non-separability so if a system is entangled then one can't talk about the states of its components independently entanglement is in in that sense is a limiting case in which the interaction across the boundary is both strong enough correlated enough that one can no longer distinguish

between the two interacting systems so that's a that's a very different meaning than some more classical uses of the term entanglement which just mean a very close correlation so entanglement is the word is used in physics of means a correlation that effectively destroys separate identities so if if a system is entangled uh one in a sense can't draw a meaningful boundary across it um but I I would like to say something uh in response to what Carl just said and in your question about you know since real real Consciousness uh experience right real phenomenal it clearly depends on um I mean organisms or systems in general can clearly vary enormously in what they can detect about the world and how they can act on the world for example you could ask for a system uh can it detect spatial dimensionality in the world you know does it live in a world that that appears to it to have any spatial coordinates and clearly for our kind of Consciousness we have to be able to to Overlay some spatial coordinate system on our experience or it doesn't make any sense at all we can't talk about object identity we can't talk about motion uh we can't talk about any of that stuff without spatial coordinates and we can ask can a system detect the passage of time which requires having a certain kind of memory so that I can tell now from what happened you know a few seconds ago that depends on my being able to remember a few seconds ago so these are capabilities that systems have to have to have experiences like ours but I think of a dividing line that I might like to propose is between systems that are capable of acting in order to get more information in the future which which Carl has characterized in his work as acting in pursuit of epistemic value uh in an ordinary language when one would characterize this curiosity acting with curiosity and I think that uh what is characteristic of the systems that we tend to regard as having phenomenal consciousness of the sort that we have is that they act with curiosity fairly clearly other mammals for example are all curious so I would just throw that into the conversation as a as a a thing that I think characterizes the sorts of systems that we regard as interesting from the point of view of what their experience is like you know and just to um bring the last two points together in acting in a way to resolve your expected surprise which is just uncertainty that would be the definition of curiosity you have to have um a frame of reference a structure on the inside that can predict the consequences of any particular action um so by definition this introduces a form of temporality you know whether the agent knows that this is time uh but there's certainly an ordinal aspect that something will happen contingent upon doing this and I think that This lends a um well some people refer to the temporal thickness or temporal depth to the way that we make sense of data and I think that that comes for free um when you talk about systems that have this uh this kind of curiosity curious creatures um that I think you

you would have to entertain in order to uh to um as um requisite for having any kind of phenomenal Consciousness the reason I say that of course is that that um modeling the consequences of action over some time like um and sequence just as memory so you know I think the two come for free from the same commitment to certain systems that have this capacity to not only resolve or minimize their surprise in the moment but also act upon a world in a way that they anticipate that they will minimize their expected surprise and resolve uncertainty and it is something that has underwritten all of our behavior and certainly behavior in statistics since since the 1950s so this was um this is called the principle of optimal ex Bayesian experimental design namely what is the best experiment what's the best action what's the best sort of writing to the world acting upon um that will minimize my uncertainty and that's the same as maximizing my expected Information Gain that you can compute so if you know the state of the world do you know what you don't know then you can estimate how much uncertainty you can resolve if I did that as opposed to this and of course that's just what we do as scientists we design our experiments exactly in the way that resolve the greatest amount of uncertainty about our competing hypotheses as scored by maximizing expected Information Gain it's something that re-emerged in the 1990s in the guys in machine learning is active learning what data point do I commit to trying to search for that's going to best resolve the uncertainty about the unknowns in my particular World moral or statistical model of of these data so I think what Christmas just said is absolutely uh absolutely Central and notice that there are probably very few of these kinds of systems in our universe um you know a thermostat doesn't have memory a thermostat but very simply cannot plan what we're talking about here are systems that at least look as if they are panning so there's a whole universe of very functional very adaptive systems creatures particles phenotypes out there that don't plan the weather Evolution um thermostats um but there may be a small number of things that do have this these temporary thick World models under the hood the structures that we were talking about before that enable them to invince this kind of curiosity and it was exactly that kind of action I was I was sort of hinting at before but I didn't think we were going to get quite so foreign into the weeds in terms of that's exactly that's my brain how it function yeah I'm sorry the don't fear it because it's how my brain went something stuck I need to get it out so we quickly move on you know when when Chris say observers is everywhere observation is an information transfer Chris say that so if you have to Define what an observer is could you define it precisely what an observer through a physics science and Kyle could you define on the Neuroscience site because to me Observer is the Observer has information am I correct it

could be looking at like the ego or the identity but Chris says observers are everywhere so can we just unpack The Observer that what both of you mean could Chris sure uh when I talk about an observer I admittedly use the term very loosely in a way that I think lots of physicists do when they see things say things like The Observer is just another quantum system I think of an observer as minimally uh any system that interacts with an environment such that the joint Observer environment System state is separable so they're not entangled uh and under those conditions one can talk about the state of the Observer and how the state of the Observer changes as it receives information from the environment and as it acts on the environment so an observer is something that can receive information and act in a conditionally independent way from its environment now that's an extremely minimal definition and lots of things qualify so one can now start layering abilities on top of this to make it more and more capable so when we use the word observer in English we're referring to systems that at least can pick out different objects in the environment and act on them reasonably specifically and I say reasonably specifically because there are no actions without side effects uh but can act reasonably specifically on objects that it could distinguish from the rest of what's going on in its so this gets back to our previous conversation about temporal depths and object identity and all of these other things there are these two senses of Observer one of which is extremely minimal and only really depends on the existence of a Markov blanket which is what separability gives you and the other which can be arbitrarily sophisticated uh with with large a large memory for example a large ability to model statistically uh the data that it receives and an ability to act with a fair degree of precision on the environment that it sees and I would say everything in between as an observer with some set of capabilities or other they may be fairly small and they may be quite sophisticated Kyle um yeah again I agree entirely um I mean in my world observation which is the thing that defines something as an observer measurement inference and existence are all the same thing um so again from a different inflationary taking the deflationary perspective that Chris has articulated and anything that can be separated or identified individuated from its embedding environment must by definition have a Markov blanket that Markov blanket must comprise some sensory States so it must be um its existence belies the fact that is evidence that it is observing In This Very minimalist um sense that it is processing sensory and I would contend that you know this can be read I repeat as measurement it can be repeated you're in the sort of quantum or physicists sense and that very um minimal definition of what it is to be an observer it is to be an observer is just to exist from the point of even Neuroscience you're a scientist um I think you're talking about this more

elaborate kind of observation that has this curiosity attached to it um so the kinds of particles that I spend my time measuring um our people and brains um so these these particles these very complicated very expressive particles um have a particular kind of observation that basically we were talking about before which in which is um the kind of observation that they can plan to make the good observations the uncertainty resolving observations the um the just is an expression of this responding to epistemic affordances responding to Salient cues in the um um in third queues in the environment to show this um curiosity that characterized so if you limit um sort of observation to this informed and observation then I think then I would then that that would be the kind of Observer that uh that I would study um as a neuroscientist is that like is that a sufficient definition um you know I do repeat that the notion of an observer is it's almost unnecessary um in the sense that as soon as you can separate anything from everything else the coupling between everything else and the thing in question constitutes an effect a kind of observation you know then the interesting questions are what happens on the inside of the thinking question uh that's basically the story that the fep Tells in terms of trying to describe this information processing this belief updating in relation to things like Bayesian inference and sense making and minimizing minimizing prediction error and ultimately planning to minimize prediction error by taking the right kinds of actions designing the right experiments which is exactly what we're doing now yeah so you know you you asking me questions and me and me trying to work out what's going to resolve your uncertainty uh this is exactly that's kind of curious behavior that I think it characterizes the kind of observers I think that you're talking about when it comes to getting a response from a neuroscientist yeah yes have you heard anything to add on no I think that that sums it up quite nicely I I I really like the the the motto uh to be an observer is just to exist yes I completely agree so much and David Bowen say The Observer is the observed so what's your thought on that then The Observers say The Observer is the observed it's like when we talk about the inside of it right you have an observer position you're having a new goal so you can observe yourself that's what they're talking about right um so I just wonder what's your thoughts on that right because your Observer is just existing whereas if we can go deeper The Observer is just observing with no content that's what I actually thought it is so I'm still a bit confused on that but yeah yeah so I I think that's a great question that could be taken in two directions One Direction which I think we won't go down is that there is a simple truth that whenever there is an observer there is also an observed in a simple sense that we're talking about before if you have any universe and you carve it into two you put a partition or a boundary between one part of the universe

and the other part of the universe and then one becomes an observer but there's also something on the other side that is um is observing The Observer and that phrase observing The Observer is a good description of me as a neuroscientist spending most of my life trying to work out what's going on inside other people's heads and it's you know it's actually a very difficult problem um but I think you're talking no now more about me observing my information processing in the spirit of things like high order thought theories of Consciousness or metacognition um perhaps even touching upon the Notions of self-modeling um speaking here on the work of people like Thomas Metzinger and I think that that um that's an essential part of the kind of observers that we're talking about when we're talking about people uh and possibly you know most animals that yeah I think probably most animals um and um in a sense that that sort of um observing The Observer uh The Observer becoming the observed is a necessary consequence of something that you actually said in your question which is the notion of depth so depth can certainly be applied in the temporal domain but also has an important um it is an important attribute of the structure that we use to do our information processing and our belief updating the structures I'm referring to structure here in the spirit of structure learning a world model or an internal model of what caused my Sensations now if you have a deep structure then by definition you have a hierarchical depth and if you have a hierarchical depth each hierarchy is observing what's going on at the level below and so the very existence of systems that whose bulk or internal states are doing the kind of information processing that enable them to plan of a hierarchical structure they have both a temporal and hierarchical depth and therefore they they must if you like have this attribute that you can read them as observing themselves simply because they possess this hierarchical structure just to finish um that notion of deep is exactly what underlies deep learning in machine learning so and when when you look at the Fantastic achievements of sense making and sort of recapitulating saying generative artificial intelligence generative AI it invariably involves a deep neural network or a deep World model and that deepness just depends upon having hidden layer upon hidden layer upon hidden layer that defines this hierarchical structure that's sort of frames of reference and frames of reference of frames of reference usually increasing in time as you get deeper and deeper into the hierarchy or in some instances a heterarchy so I think there's there is literally a deep truth to the notion that the kind of sense making and the kind of um the way that you and I experience our world um is um deep in the sense that it can be read as a hierarchical sense making where each um each level is both observed and observer it is observing the level below and it is observed by the level by the level above thank you Chris well I I think that's uh

excuse me a very nice way of putting it I I would refer to some terminology that our our colleague Mike Levin has proposed of a what he calls a multi-scale competency architecture which all organisms have each each level of the hierarchy is competent to do certain things essentially on its own without being told and it's that sense of hierarchy where at each level there are certain defined capabilities that is at play here in these kinds of hierarchical models um but but I'd also go back to another way of addressing your question in terms of self-observation and and ask what's the minimal kind of self observation that's called for in this kind of theory and in a sense it's it's some kind of observation of what is my stress level you know what what is my current level of uncertainty um and one can see this as far down in phylogeny as one chooses to look so for example molecular biologists very early studied one of the earliest studied systems in microbes like E coli is the stress response system and it was first identified by studying responses to heat stress so the whole family of response genes got called heat shock genes but they're really stress detectors any any cell even down to to prokaryotic or microbial cells uh has a a huge repertoire of of stress detectors and defensive strategies of one kind or other that don't make any sense physiologically unless we understand them as measuring their own state as being able to detect that they're being attacked by viruses or uh their temperature is high enough that their proteins are starting to fall apart and quit working or they're too cold the opposite extreme and and diffusion is no longer working and so reactions are slowing down so this sense of self-observation if you will is is very very deep biologically and seems to be a a defining feature of the things that we refer to as alive it's just the awareness of it right Chris uh well yeah I don't think that the term observe makes any sense without thinking of being aware of what's one of one is observant yeah so yeah I think I think that the the poor bacteria actually don't like it when you heat them up too much yes um just thinking about that question from the point of view or that point about uh stress detectors um on the point of view of Psychiatry um and the sort of The Human Experience I think there's um another important uh truism here we've referred to or Chris has referred to repeatedly The Importance of Being precise the importance of rendering everything predictable and unambiguous as possible you know come in one way of reading this imperative that underlies curiosity is to resolve uncertainty and keeping things very very precise if if that's the game and um I would suggest it is the game in this in in a you know in a very mathematical sense that you know if I'm in the game of minimizing the surprise I expect following this action then I am simply trying to minimize the expected surprisal which is the entropy or the uncertainty I'm always trying to resolve uncertainty about my lived world if I

was able to measure the uncertainty of my current state of being then you can see the increases in uncertainty uh are going to be have a very negative valence and decreases of uncertainty the kinds of things that I'm going to Aspire to and in one sense of course novelty seeking information seeking is exactly to reduce that um that that uncertainty but the converse um I think holds the key to many mechanical understandings of Psychopathology and anxiety so usually in Psychiatry when you look at the the sorts of aberrant belief updating and sense making that underlie all sorts of uh um conditions um ranging from a professional compulsive disorder through to autism schizophrenia through to um depression um then you can always find uh and not least of course anxiety anxiety disorders particularly generalized anxiety disorders and and sort of chronic stress all there's one common theme throughout and that's a failure to resolve uncertainty um a failure to eliminate in the right way the uncertainty that you sense yourself as currently experiencing there are all sorts of ways of resolving uncertainty you could just hide from the world so you could have agrophobia or you could become socially and emotionally withdrawn um and even physically you know manifest psychometric symptoms of and severe depression um if you had severe autism uh one way of minimizing your uncertainty and surprises is to avoid any potential situation in which was so interpersonal interactions in which there's going to be unpredictable or surprising responses or indeed create your own predictable Sensations just your skimming Behavior self-stimulation behavior and I think all of these um sort of pathological although strictly speaking there is no pathological Behavior it's all baze Optimum at some point um but these aberrant kinds of behaviors you know if the person was like me um can always be read as an attempt to minimize that the kind of stress that this is stress which can be read literally in terms of the entropy of the uncertainty associated with particular belief States during during your sense making so I think that that perspective uh has a lot of tactical utility when it comes to formalizing theories of bad sense making uh in Psychopathology and indeed as you know has found a certain Traction in feels like computational Psychiatry thank you we're going to go straight to Andy your question Andy who your question choose oh okay wow um thank you so much guys this is amazing um my pre-prepared question I think has just dissolved um so so I'm think of this more just a participation of the conversation in you guys Minds is and this is for either of you is an understanding of our Consciousness um dependent on some notion of Consciousness being a fundamental thing um separate from our sort of physicalists understanding of the universe you know like of you know complex energetic information processes arising from the standard model and plus whatever sensible near future extensions to that model might happen like like

necessarily fail in describing our Consciousness um that's as good as I think thank you so much Carl do you want to take that first yeah I think that that question speaks to yeah what was quite a large part of the focus of our conversation which I think is the distinction between um the Consciousness that has a phenomenal aspect uh and usually how the self-referential and perspective perspectival aspect to it that can be articulated not least to ourselves it include some kind of self-modeling so I would I would just fall back on um this you know the way that we would describe the kinds of systems that can support that kind of information processing and modeling and I repeat sort of modeling um oneself as part of the active process of gathering the data that you are trying to trying to model so again we come back to this notion of soliciting the right kind of information the right kind of cues and feedback from the world from nature from our from our partners and our and our friends and colleagues um uh which is this sort of curiosity you're planning to resolve uncertainty and I I repeat you know that is the special Preserve of a very small number of systems possibly just the handful of people we've found we find on um or animals that we find on on this Earth and in that sense um a description of um measurement and um the physics of measurement a metrological approach to physics would not be sufficient I think to fully understand that kind of Rich thick deep curious Consciousness that has this phenomenal um phonological attributes um and I'll just um I also just put out there for Chris's comment um you could argue that standard the standard model and that physics and all the mathematical and narrative languages upon which that is predicated is just a shared construct it's just part of our world model our sense making that actually inherits from a kind of collective unconscious that would be ascribed to um evolutionary psychology or cultural Niche construction but at the end of the day it's just another hypothesis maths is just another hypothesis that we use to make sense it feels really solid because it's shared so universally and has that unreasonable decision and efficacy just to come back to a really key um point that Chris has made repeatedly if it if it's precise in its explanatory power then it's going to find a home in our existential architectures the structures of the way that we make sense or anything that survives but in particular the way that we make sense of our world Chris over to you for that one yeah yeah thank you all right I I would just add that uh you know the the standard model is a lovely construct uh it's a theory of interactions and the symmetries of interactions it's it's not a theory of what exists and uh essentially what we've been talking about today is a way of Shifting our thinking about uh what systems are what it means to be a system that's interacting with an environment and when one talks about information exchange or information or experience of ones using a

language that is not captured strictly speaking by uh a standard model like description of the symmetries of interaction because one's added to that description of some description of the things that are interacting and that's uh in in a sense has been provided by physics with the notion of a particle or the notion of a field but even underlying those notions uh is this very informal notion of a system and it's that notion of of a system that I think has to be added in order to uh fully capture what we've been trying to talk about today I would say that that what we've been discussing today is mainly a different point of view on the physics of interaction than has typically been pursued in physics up until I would say the past couple of decades where a lot of attention has gone to describing physical interactions in terms of communication protocols and information exchange using some of the language that we've been using today so I I don't think that one has to assume if you will new metaphysics I think when uh just to take just has to take a slightly different point of view on sorry we're breaking in Andy can we move on quickly to Kyle please wonderful and um yeah I invited people to the stage just to tweak the algorithm and bring um more years um to the um outstanding amazing speakers here and thank you both very much for being here um a couple of things that I was listening to that I took uh particularly interest in first was like the anticipation and the modeling I guess I've uh been in sports activities and careers in which uh I've had to have a great deal of anticipation um so much so that I I'm almost thinking about the other um aspect of it because uh when there is no surprise people become kind of conform they conform to the uh conceptual understanding of um of the system itself and they lose uh spontaneity so I was just thinking about the benefit of uh you stress because address was mentioned before but there is a distress in eustress and spontaneity for surprise another quick um and so I was just wondering if you could just briefly touch upon uh spontaneity Carl and then for the both of you I was just wondering um I've been hearing some stuff about like collecting data for a representation of self and I think this is what marketers do in companies that are taking all of our data they're kind of like creating a digital twin to anticipate how we will actively infer throughout our lives and taking advantage of that I was wondering if you have anything to say about that thank you I'll take the spontaneity question which is a really interesting question um and I one could read that in one of two ways so um in terms of uh say sports physiology or um sort of the the odd the agenda of uh becoming a good sports person um spontaneity may be read in terms of automatization and habitization becoming fluent you know getting into the flow in the particular situation in which you have become skilled and and I think that would just be an expression of this imperative to um do actively very precise belief updating uh with

um resolution of all reducible uncertainties in exactly what you're going to do um and effectively you are responding reflexively and automatically in the situation um through a series of movements or responses over which you have Exquisite control and also that you can predict um expertly so this would be if you like the end point of an uncertainty reducing um scheme but there's another kind of uh if you like spontaneity that is afforded um when we now turn to the construction of deeper and deeper models so you know when one tries to model this kind of process one then thinks okay well what if we now put a deeper model or a deeper layer on top of the lower layer that's doing the spontaneous automated behavior usually that there has a more temporarily extended aspect to it um and um and now contextualize um uh and make sense of um different the evolving of say a game at a a uh at a much longer time scale um what can happen with some uh you know once was established these um deep generative models or World models of me performing in a particular context is actually about apology where you start to go back and start to attend to your more fluent ultimate uh automatized behaviors and that can completely destroy the behavior and so I have colleagues who spend a lot of their time in clinics um neurologists uh dealing with functional medical symptoms that arise when a sports professional can no longer play tennis or putt simply because they're focusing they're attending to the things that had previously been learned over learned and and become reflexive so this is one of those nice instances of getting the estimates of uncertainty and precision wrong in your belief updating when you're actively engaged with the world the final kind of spontaneity I I guess you're hinting at is really that that underwrites creativity and the way that we can spontaneously explore hypothesis spaces um that provide alternative explanations for our sensed world or indeed the world that we can create in terms of um being artists or poets or uh inventive in our personal relationships so that that that that um I think that's a very different kind of spontaneity and I think it speaks to this impa this aspect of curiosity there's imperative to resolve uncertainty but if you want to resolve uncertainty you have to have the capacity to develop a state of alternative hypothesis hypotheses over which you can then resolve uncertainty so um I would look at this uh as a one fundamental aspect of novelty seeking having spontaneous ideas aha moments that just um Encompass different explanations officials for the way the world works and then you once you have this hypothesis space then you're able to either create or secure evidence oh it works like that this is a you know a nice simple and Central resolving explanation well I'll I'll just make a remark that uh when you talk about use stress uh in a sense in a sense you're talking about play which is again something we observe in in all of the organisms that we think of as as having interesting sorts

of so yeah I I agree that it goes back to curiosity but yeah thank you those are organisms that can have fun thank you Carla for that great questions can we move to prakash quickly yes thank you Ivy and um you know thank you very much both the speakers very deep dive into what we think of as interaction between uh Observer and The observed or what we call as Consciousness in the material around it I want to overstep the boundary a little bit and go deeper and ask if you're having a kind of spiritual experience or if you are under the influence of psychedelics what what is the state of the Observer there is the is there a witnessing that happens without an observer and I just want to explore a little bit on that side because it is an altered state of consciousness so I just want to see what opinions you have on that uh sure I'll say a little bit about that I think one characteristic of these sorts of experiences is an absence of the usual kind of memory encoding which has a has two consequences I mean one consequence is that one doesn't remember exactly what happened uh and that's true of much of ordinary life as well but uh another consequence is it allows experience without a whole lot of processing we're we're typically doing a tremendous amount of processing thinking about the information that we're getting uh thinking not necessarily consciously but uh we're doing a lot with the information that we're getting and uh using it to plan ahead to uh adjust the overall structure of our network of expectations about the world and if you turn that off of I think you get to see more in a sense of what's going on because uh one isn't having to expend the energy to do all of the processing that we're constantly doing so we don't this this also applies I think in at least some sense to the dream stage which is also an alternate state of consciousness one doesn't have to spend a lot of effort for example trying to figure out whether the object you're seeing now is the same object that you were saying a little while ago oh so it's okay if that object is changing its its properties in some uh interesting way so I think these experiences are very interesting uh it is kind of clear why in a sense they're uh liberating or can achieve a certain level of of house cleaning in the mind by stripping away expectations and not doing the kinds of uh cognitive work that we're often doing to cut experience up into little manageable fragments because all of that depends on being able to keep a lot in memory and to devote a whole lot of energy to computation so backing off from that uh seems like clearly something that can be valuable say that again Chris I said you're probably more familiar with these sorts of experiences uh in terms of their neurological and also phenomenological properties absolutely to be clear Chris does not mean that I take lots of drugs it's because I'm a psychiatrist [Laughter] so again that's a great question Ivy you're going to be very cross with me because I've got so much to say about that question because I'll say it very very quickly um

so there is a very well worked out literature addressing exactly this question uh and some of the words that Chris used in answer um figure very prominently in this literature and the literature has two sides and I'll just start with the notion of um house cleaning and housekeeping and the important insights one gets from sleep Research into our constructive realities and how sensitively they did upon um the Precision of our belief updating so in the brain the Precision of belief updating is literally described um in predictive coding formulations of this kind of information processing in terms of precision weighted prediction errors and the Precision is usually set physiologically by neuromodulatory transmitter systems things like 5H serotonin 5-HT noradrenaline dopamine oxytocin amino-like and and the key inside here is that dreaming as an altered state of consciousness is principally determined by the status and discharges of these ascending neuromodulated transmitter systems so there's a very concrete very well understood empirically set of mechanisms that tie together the Precision of your belief updating and the um the differences in the way that we make sense of our world even when asleep you know when we are dreaming we are still making sense of a world it's actually our self-constructed world that is sequestered from the actual sensory input by turning down the Precision of the sensory prediction error parents but not reducing the Precision of all the hierarchically deployed prediction errors in our brain um so dreaming and just the existence of dreams I think is as my um as well colleague who died a few years ago and Hobson one of the world's experts on sleep and dreaming would say if you want evidence for the brain as a constructive organ making sense of its sensory Impressions on its holographic screen or its Markov blanket you need to look no further than the fact that we can dream that is the only reality we really need to uh to worry about but it does depend you know the the kind of sense making does depend very very um sensitively on neuromodulation that controls the Precision just one little link here because Chris also used the word temperature in my world Precision is inverse temperature um so you talk about soft matte premises in those temperature parameters that is precision so we're talking about cool-headed sense making um now how does that relate to the other literature well the other literature is on waitful sense making where you interfere with exactly the same neurotransmitted systems so all psychedelics effectively act as agonists of private to 2A receptors which are really important receptor that's set in the Precision or the gain or the Precision weighting of the prediction errors and that to a certain extent can be thought of as the inverse of sleeping so here you're emphasizing the Precision of lower level sensory prediction errors with respect to your higher deeper kinds of constructs so I think um the question was absolutely right that that you know

by pharmacologically altering the Precision of our belief updating on the temperature differentially at different hierarchy levels of deep sense making can have profound effects on the way that we exert control or put prior constraints or pry beliefs in our experience into the mix when making sense of our Sensations I was just going to conclude by um pointing to the literature on psychedelics um the key or one of the key authors in this field is a gentleman old Robin Carhart Ares and the bottom line of the story that he tells is that psychedelics reduce the Precision of very deep prior beliefs and enhance the Precision of more sensory level processing and that basically means that you are dissolving or relaxing the Precision of higher level higher level beliefs so you are becoming more attentive to the sensorium are we going to quickly go to re yeah so uh the thing so it's kind of a follow-up question to what prakash asks then it is like you know if there is a sense of Observer not being there then how can how can that be you know experience be valid or the Consciousness be valid on that and of and the reason why I say that it is a follow-up question because uh because the moment you we don't that's what Kyle said just now we don't have that distinction of you know observable is observed and we don't go into that distinction as much and then just factor in of conceivability with the concept of Markov blanket as how much that conceivability can exist and beyond that if there is what Kyle is referring as Dark Void so if it is audible uh you can you guys can you know as prakash is giving thumbs up yeah you can you guys can you know answer on that I'll try to say something here uh I I see this question as asking about the limit of weak interaction between two systems so if you think about a boundary between two systems that's supporting some interaction one can always think of that as a Communication channel from a formal perspective and someone can ask about the limit as the bandwidth that that channel goes to zero uh and as the bandwidth of the channel goes to zero there's no information exchange so there is in a sense of no experiential content anymore and I suspect that these experiences oh that you're describing are experiences of uh approaching that limit of no information exchange across the boundary I mean one one can there's a kind of Brute Force way to approach this limit is something like uh using sensory deprivation but of course that doesn't approach it very closely because it doesn't remove uh interception so I think uh I think it's a it's a very difficult limit to actually approach and and remain alive um but I think one can get closer than sensory deprivation yeah see you can also argue that um certain sleep States approach that in terms of neurochemical sensory deprivation basically switching off apart from um afference from the ocular motor system which are part of the central nervous system everything else is basically switched off so you um you met one may be asking now well okay if there's no information

traffic across my uh Markov blanket um how do I make sense and I think the the answer to that comes again um from these this notion of a deeply structured internal uh model that of course is defined by a whole series of Markov blankets inside the head so it comes back to observing The Observer um that there's still good information processing you can do even the absence of Sensations from the sensory receptors that you know people have um and there's an interesting mathematical take on this which comes back to business notion of housekeeping um if you remember right at the beginning we were talking about the imperatives for or natural descriptions of things that exist inside and with Markov blankets which is to maximize the um evidence for effectively their internal models evidence can always be written down as accuracy minus complexity which means there is still evidence maximizing information processing that can occur in the absence of sensory inputs which can minimize the complexity and what that translates to in sleep is basically exactly the the housekeeping removing redundant parts of the structure as we rehearse and find out what associations that we built up during the day are no longer necessary to provide a simple minimally complex but accurate account of what we have seen to date but this only works because I you know there are Markov blankets within Markov blankets screens within screens within screen so it will only work with um deep generative models a thermostat will not be able to Dream Will you and I can so uh but just just connect it to the previous question as well um I'm I don't know anything about that you're the Buddhist um constructs that would be useful to yeah to um to complement this perspective but it does strike me that um the action of psychedelics um is in many senses trying to Aspire to states of mindfulness which you know where if you're a skilled meditator that basically means you now have some volitional control over the attention of the Precision that you afford certain sensory Sensations say it's by breathing or you know another kind of modality which of course is exactly switching off the Precision of the deeper higher internal uh structures which would be in one sense um a dissolution of selfhood um that is the you know the um red as having higher beliefs and constraints in the game when making sense of your Sensations that would be if you like the aspirational target of ultimate kinds of mindfulness so I think there is a deep connection you know between the role of psychedelics and pharmacologically biasing your sense making to very precise low-level sensory inputs um and the same ambition of mindful states that are you know that are the Holy Grail of certain kinds kinds of meditation the final point I want to make in relation to or through the lens of Psychiatry is that there are some very interesting out-of-body experiences and derealization depersonalization experiences that people have that can be accessed through the use of

psychedelics and one of them is the sense that I am not me or I am not even a person um and this kind of depersonalization can be truly awful if your subject to it um and what it tells you is that the hypothesis I am me is just another construct and then it doesn't necessarily always have to be in play when you're making sense of your Sensations and we'd sort of take it for granted um that we it's me sensing and having these experiences but that's just part of the self model and it is just a model it's just a hypothesis that we use to make sense of things and there's no reason why you can't explore alternative hypotheses that I am not me and sometimes if you commit to that and that provides great it is a better explanation for these very precise prediction errors that may be pharmacologically or pathologically um precise um then you will experience um a kind of nihilism of a deeply disturbing sort where you no longer sense that you are you and there are all sorts of interesting symptoms like Capgras symptoms where that same kind of um identification um is transplanted to other people so my wife no longer is actually my wife she becomes an imposter um so a lot of things we take for granted in terms of selfhood and self-awareness can be very very fragile um some psychiatric conditions or under some uh under some uh under some under some drugs but you know I don't know whether that's the kind of Buddhist perspective of the question that was uh was chasing just for your benefit absolutely in chapter 27 of the diamond Sutra it uh warrants against nihilism just for this exact uh point that you brought up uh so intuitively so thank you thank you we've got to move on now we have only seven more minutes uh we're gonna move on to John and then maroon and then that was it it's going to be end of it because we only got seven more minutes well thanks Ivy I'll make this uh relatively brief I I'm thinking of telling my wife that she's an imposter so that I don't so she doesn't whisk me off to the desert as um at any rate this has been great I'm kind of a I'm a novice physicist in all this I'm only exploited by ivy for my my audio engineering skills which I have so on display today but I wanted to just I have lots of questions but I just want to one that's occurred to me over and over and it's been something that's been nagging at me for some time is this idea you know I people talk about the hard conscious of hard problem of Consciousness uh qualia and all of that but it seems to me that the harder problem really is about curiosity what you guys were talking about curiosity I use I always use the word creativity I try to imagine for me artificial intelligence will really be artificial intelligence if and only if it gets to the point where it can look out at the stars and say wow what is that um maybe we could design a rocket ship and put things on it to go out there and take a look uh you know or collect samples and bring them back and I that strikes me as a really hard barrier creativity and it's lesser versions of that perhaps are in the Arts and

everything else but anything having to do with creativity strikes me as the really hard problem of Consciousness and I wonder if you think artificial intelligence will actually get us to something will artificial intelligence be that good at any point is that even possible stump them I I was going to suggest that that you could start with uh all right yes yeah it does strike me as a question probably more for Carl especially since that's what he's supposed to be doing right no absolutely I think what you've just said is absolutely spot on um and I I think that um the Next Generation Um of artificial intelligence you know it's probably closer than you might think and it does exactly rest upon this curiosity so when Siri starts asking you questions as opposed to you asking it questions then we'll know we're getting much much closer so I think you've identified exactly the next moment you know the next gtp3 GTP uh moment it's going to be when these schemes are not just generating content for us that actually want to know about us and how the world works and then start asking questions that has this that has this um aspect of curiosity this response to epistemic affordance is that you are absolutely right you know um is one expression of creativity and I'm just mindful of uh one of my friends Jurgen Smith who the inventor of the long short term memory STM that is in your mobile phones probably at the very moment he's always um played um a or told a story which um produces a very close link between the um that complexity minimization that evidence maximization or free energy minimization that we were talking about before which when read in terms of information Series right is called compression so the key role of compression in terms of getting simple but accurate accounts as simple as possible no simpler accounts of our sensorium but that manifests in terms of behavior and actively going out there and doing smart data mining it manifests as curiosity it manifests literally in his word as artificial thumb so we come back to what Chris was saying before about playing play is itself um one expression of this uncertainty reducing imperative yo whether it's children playing together or you playfully asking me with ridiculous questions or whether it is you know the kind of play um the kind of curiosity um that um so yes I've witted on for two for too long but to my mind fun creative creativity and curiosity um are all expresses of exactly the next thing that we'd expect to see from our intelligent artifacts very good Moran yeah um thank you very much uh to both of our guests um as well as the panel members um I just had a I have a few questions but I'm just going to narrow it down to one and it is about one of the statements that was mentioned uh to be an observer is to exist I would have thought that to be in order to be an observer at the very least you have to exist right so because for example a book exists right but and because it exists you can think about a Markov blanket around it but again having said that there is a

Markov blanket around it like if you want to say that it's observing because it exists then you can think about like I don't know like imagine a sense serious State and an action state but because it doesn't have cell food it can be the sensory set would be to be read to be thrown around for example in the past in a completely passive sense and the action state of a book that doesn't have cell food could be palpating my my eyeballs so that I could like read it like you know like it it's not as a certain state of like a weird sort of action selection mechanism so just want to hear your thoughts um like if there's time both Professor first and then Professor feels that would it make sense to say that in order to be Observer at the very least you have to exist and plus other conditions yeah if I could start I I think that's very fair I I think the point that Carl was making with that slogan uh and a point with which I agree is that if something exists in the sense of it it has a stable and weak interaction with its environment so so if if we are outside Observers we can see that thing and say oh that's an object that's a thing that's stable in time that has persistence if a thing is stable in time and it persists then it is interacting with its and that interaction is two-way um and it's interact the interaction we may regard it just as thermal right we we may just see it as heat exchange but it's still an interaction and so this very minimal sense of observation that we've been discussing is just interaction it's just physical interaction by systems that are the systems that have a sufficiently stable go ahead hello no I was just going to say that's that's that's what we mean by this slogan that existence is interaction interaction is observation uh just to reinforce that point um and also celebrate the the notion of a book I've never been asked that before that's an excellent an excellent example usually people ask about Stones Chris's observation that the stone is actually observing the temperature of the ambient environment thank you very much and this is a very Elemental basil kind of observation and which is you know that's not the kind of observation we're obsession with curious breached with these deep splotches but it's still observation from the point of view um from one point of view but the book is is even more interesting because because it has a language in it it is actually observing human culture of the human race so very much like desire paths in Niche construction things that basically exist and can be read as the environment learning about its inhabitants behind the environments Markov blanket you can actually read the printing of the book and the fact it was made by human beings but also the content in the form of the book in and of itself as one way of observing its environment but probably best read as learning about its existence is it it owes its existence due to the printed word so it is learning about you now I know it's not observation in the moment but at a certain time scale it's a kind of thank you thank you very much won't ask you any questions so it's you

don't have to worry about it I think we wind down to the internet or both of your time so can we all give a massive thank you to Professor Friston with us today and Dr Phil today please everyone yeah it's not expensive yes and thanks to you all for for interesting questions happy Lunar New Year thank you for your interesting answer thank you yeah good thank you in conversation thank you so much for both great mornings thank you hope to see you guys thank you very much ciao thank you for organizing this question a really profound discussion in the question and the session which I think if anybody is interested should listen to the replay a few times because they've been very deep into the architecture what Consciousness is put in words it's not put in words in books this came out exactly through an exchange of conversations and this is what this kind of discussion yes just to be cleared for people in the clubhouse room we are still sitting here and and uh both Dr Friston in fields are are listening in so you know if you want to like sneak in some uh you know nice comments that would be great too but I think we are going to wind down the zoom meeting because because I have to go sorry you guys